Operating Systems: Study Notes

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1. Page Replacement and Page Faults

Virtual memory lets processes use an address space larger than physical RAM. A **page fault** occurs when a process refers to a page that is not currently in physical memory. The operating system fetches the page from secondary storage. If every available frame is occupied, it must choose a victim page to evict using a page replacement algorithm.

2. Least Recently Used (LRU)

Least Recently Used (LRU) replaces the page that has not been accessed for the longest period of time. It uses the history of recent page references, based on the assumption that pages used recently are more likely to be used again soon.

Example: Suppose three frames contain pages A, B and C. Their most recent access times, from oldest to newest, are A, C, B. When a new page D must be loaded, LRU evicts **A** because A was accessed least recently. The resulting set of pages is B, C and D.

An LRU implementation may maintain timestamps or a data structure that records access order. Accurate LRU can add bookkeeping overhead because the access history needs regular updating. LRU aims to exploit temporal locality: a page accessed recently is often accessed again in the near future.

3. First In, First Out (FIFO)

First In, First Out (FIFO) replaces the page that has been in memory the longest, regardless of when it was last accessed. FIFO tracks arrival order, not access recency. Because arrival order is simple to track, FIFO is easy to implement, but it can remove a frequently used page merely because that page arrived earliest.

4. Comparing LRU and FIFO

Property	LRU	FIFO
Victim page	Least recently accessed	Oldest loaded page
Information used	Recent access history	Arrival order
Typical complexity	More bookkeeping	Simpler queue-based tracking

5. Optimal Page Replacement

The **Optimal** page replacement algorithm evicts the page whose next use lies farthest in the future. It achieves the fewest possible page faults for a known sequence of future page references. Because a real operating system generally cannot know future references, Optimal replacement is primarily a theoretical benchmark for comparing practical algorithms such as LRU and FIFO.

6. Thrashing and Locality

Thrashing occurs when a system spends excessive time moving pages between RAM and secondary storage instead of performing useful computation. It can happen when processes do not have enough memory frames for their working sets. Improving frame allocation and managing the number of active processes can reduce thrashing.

Temporal locality means recently accessed data is likely to be accessed again. **Spatial locality** means nearby addresses are likely to be accessed. LRU draws on temporal locality when selecting a page for replacement.

7. Deadlocks

A **deadlock** is a situation in which a set of processes cannot proceed because each waits for a resource or event that another process in the set must provide. Four necessary conditions for deadlock are **mutual exclusion**, **hold and wait**, **no preemption**, and **circular wait**. All four conditions must hold for deadlock to occur.

Mutual exclusion: Some resource is non-shareable. **Hold and wait:** A process holds one resource while waiting for another. **No preemption:** A held resource cannot be forcibly taken away. **Circular wait:** A circular chain of processes exists in which each process waits for a resource held by the next.

8. Short Self-Check

1. Under LRU, which page is removed: the one loaded earliest or the one accessed least recently?
2. Why can FIFO remove a recently accessed page?
3. How does a page fault differ from a deadlock?
4. Why is Optimal page replacement useful for comparison but difficult to implement?
5. Name the four necessary conditions for deadlock.

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